PART A – THEORY ANSWERS

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1. Mean, Median, Mode (Customer Income)

- **Mean (Average):**
  The sum of all customer incomes divided by the total number of customers. It represents the overall average income.
- **Median:**
  The middle value when all incomes are arranged in ascending or descending order. It is useful when the data contains outliers.
- **Mode:**
  The most frequently occurring income value. It indicates the most common income group.

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2. Variance vs Standard Deviation (Loan Amount)

- **Variance:**
  Measures how much the loan amounts deviate from the mean. It is expressed in squared units.
- **Standard Deviation:**
  The square root of variance. It measures the spread of data in the same unit as the original values.

Difference:
Variance is a squared measure of dispersion, whereas standard deviation is easier to interpret because it uses the same unit as the data.

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3. Random Variable

A random variable is a numerical value that represents the outcome of a random process.

Examples from dataset:

- Loan amount assigned to a customer
- Credit score of a customer

Random variables can be:

- Discrete (e.g., number of defaults)
- Continuous (e.g., income)

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4. Conditional Probability (Loan Defaults)

Conditional probability is the probability of an event occurring given that another event has already occurred.

Example:
Probability that a customer defaults given that their credit score is less than 600.

It is useful in identifying high-risk customers in loan analysis.

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5. Bayes' Theorem (Loan Risk Analysis)

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B \mid A)$$

$$P(B \mid \neg A)$$

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \approx 0.68,\ P(B) \approx 0.25$$

P(B)=0.25P(B|A)P(A)=0.17P(A|B)~0.68Posterior = useful evidence / total evidence

Bayes' Theorem is used to update the probability of an event based on new information.

In this project, it helps calculate the probability of loan default based on factors such as credit score and income.

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6. Empirical vs Theoretical Probability

- **Empirical Probability:**
  Based on actual observed data.
  Example: Default rate calculated from the dataset.

- **Theoretical Probability:**
  Based on mathematical reasoning and assumptions.
  Example: Expected probability of default using probability formulas.

Difference:
Empirical probability is data-driven, while theoretical probability is assumption-based.

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7. Poisson Distribution

The Poisson distribution models the number of events occurring within a fixed interval of time or space.

Example:
Number of loan defaults occurring in a month.

It assumes:

- Events occur independently
- The average rate of occurrence is constant

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8. Eigenvalues and Eigenvectors (Data Analysis)

- **Eigenvalues:** Indicate the magnitude or importance of a direction in data.
- **Eigenvectors:** Represent directions in which data varies the most.

They are commonly used in Principal Component Analysis (PCA) for dimensionality reduction.

In this project, they help in reducing the number of variables while retaining important information.